

Steam Game Ratings

CS555 / Fall 2022

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Introduction

- <https://www.kaggle.com/datasets/nikdavis/steam-store-games>
- ~27k records with 18 columns
 - Name, Release Date, Developer, Publisher, Genres, Categories, Playtime, Price, etc.

positive_ratings	negative_ratings	average_playtime	median_playtime	price
Min. : 0.0	Min. : 0.0	Min. : 3.0	Min. : 3.00	Min. : 0.000
1st Qu.: 113.0	1st Qu.: 40.0	1st Qu.: 93.0	1st Qu.: 94.75	1st Qu.: 1.590
Median : 433.5	Median : 119.0	Median : 228.0	Median : 234.00	Median : 4.990
Mean : 4324.7	Mean : 886.7	Mean : 680.9	Mean : 663.90	Mean : 7.593
3rd Qu.: 1793.2	3rd Qu.: 386.0	3rd Qu.: 375.2	3rd Qu.: 382.00	3rd Qu.: 10.990
Max. :2644404.0	Max. :487076.0	Max. :190625.0	Max. :190625.00	Max. :114.990
genre2	genre3	genre4	genre5	genre6
Length:5956	Length:5956	Length:5956	Length:5956	Length:5956
Class :character	Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character	Mode :character

(190,625 hours is almost 22 years...)

Research Question

- What are the leading factors in a game being highly rated? What further information would we need for a more comprehensive study, if any?

Data Preparation

- Calculate:
 - $\text{total ratings} = \text{positive} + \text{negative}$
- Sort by most owners (original data lists it by tiers, remove the 0 – 20,000 tier)
- Remove records with less than 3 hours average or median playtime
 - Steam policy allows you to refund games with under 2 hours played
- Apply filters
 - Median and Average Playtime ≤ 1000
 - Total Ratings ≥ 5400
- Now have 496 records of medium to high activity games

	A	B	C	D	E	F	G	H	I	J
1	name	category1	average_playtime	median_playtime	price	positive_ratings	negative_ratings	pct_positive	sp	highly Rated
2	Counter-Strike: Condition Zero	Single-player	427	43	7.19	12120	1439	0.893871229	1	1
3	Half-Life 2	Single-player	691	402	7.19	67902	2419	0.965600603	1	1
4	Half-Life 2: Deathmatch	Multi-player	311	32	3.99	6020	787	0.884383723	0	1
5	Half-Life 2: Lost Coast	Single-player	46	29	0	5783	1020	0.850066147	1	1
6	Portal	Single-player	288	137	7.19	51801	1080	0.979576786	1	1
7	Warface	Multi-player	949	104	0	32751	14409	0.694465649	0	0
8	Brawlhalla	Single-player	724	146	0	73268	14489	0.834896362	1	1
9	Half-Life: Opposing Force	Single-player	624	415	3.99	5250	288	0.947995666	1	1
10	Half-Life 2: Episode One	Single-player	281	184	5.79	7908	517	0.938635015	1	1
11	Half-Life 2: Episode Two	Single-player	354	301	5.79	13902	696	0.952322236	1	1
12	Left 4 Dead	Single-player	897	278	7.19	17951	948	0.949838616	1	1
13	Call of Duty®: Modern Warfare® 2	Single-player	999	580	19.99	30006	2826	0.913925439	1	1
14	Tomb Raider	Single-player	527	475	14.99	93509	4103	0.957966234	1	1
15	Chivalry: Medieval Warfare	Multi-player	675	293	18.99	51499	10939	0.824802204	0	1
16	theHunter Classic	Single-player	163	29	0	11440	15403	0.426181872	1	0
17	A Story About My Uncle	Single-player	193	220	9.99	17924	1642	0.916078912	1	1
18	Life is Strange - Episode 1	Single-player	412	145	0	118634	4496	0.963485747	1	1
19	Dirty Bomb®	Multi-player	715	73	0	43743	10891	0.80065527	0	1
20	Infestation: The New Z	Online Multi-Player	376	73	0	13726	14201	0.491495685	0	0
21	HITMAN™ 2	Single-player	636	514	44.99	10814	1756	0.860302307	1	1
22	Alien Swarm	Single-player	371	83	0	17435	941	0.948791902	1	1
23	Psychonauts	Single-player	491	232	6.99	8204	456	0.947344111	1	1
24	Total War: MEDIEVAL II – Definitive Edition	Single-player	469	361	0	10270	605	0.944367816	1	1
25	Star Wars: Battlefront 2 (Classic, 2005)	Single-player	618	320	7.19	29149	2425	0.923196301	1	1
26	Just Cause 2	Single-player	519	272	9.99	33636	3231	0.912360648	1	1
27	BioShock Infinite	Single-player	614	519	19.99	79442	3846	0.95382288	1	1
28	Saints Row 2	Single-player	322	81	9.99	9442	2881	0.766209527	1	0

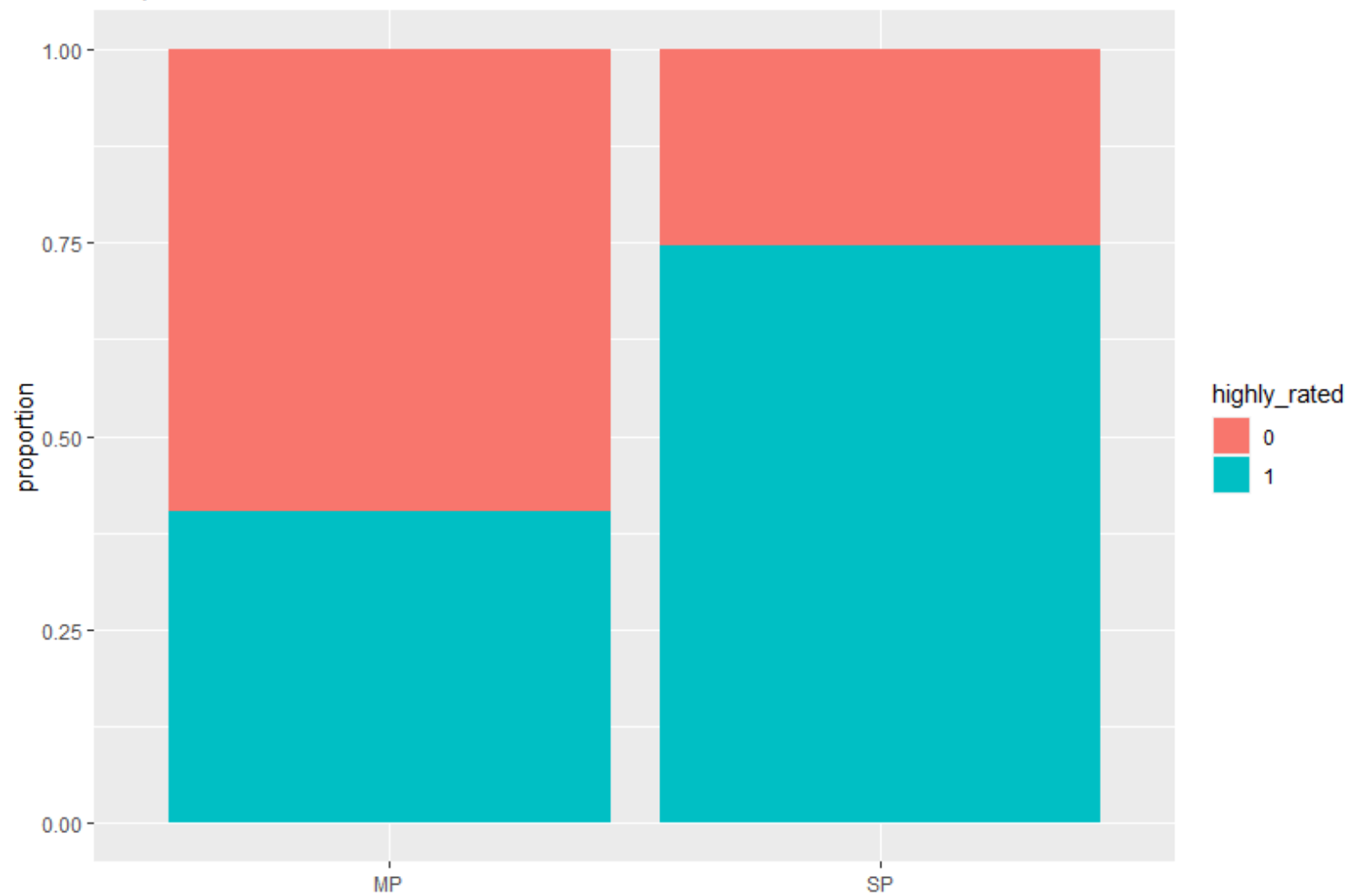
Linear Regression

- Calculate:
 - $\text{ratings_diff} = \text{positive} - \text{negative}$
- Try to predict ratings_diff based on combinations of price and playtime
- None of the model performance were satisfactory
- Calculate:
 - $\text{pos_neg_ratio} = \text{positive} / \text{negative}$
 - But this is not linear!
- After trying a few more permutations of independent variables, decided to change approach

Logistic Regression

- Calculate:
 - $\text{pct_positive} = \text{positive} / \text{total}$
 - Single-Player column = 1 if sp & 0 if mp
 - Highly Rated column = 1 if $\text{pct_positive} \geq 80\%$ & 0 otherwise
- Compare proportions of single- and multi-player games that are Highly Rated
- Found that the difference is significant at $\alpha = 0.05$
- Odds Ratio of SP:MP = 4.35

Proportion of Games with At Least 80% Positive Reviews



Conclusion

- Single-player games are indeed more highly rated by players, must investigate deeper through other grouping methods or datasets
- Playtime can be a tricky statistic in its current form, more granular would be helpful
- Factors of free+sp, free+mp, paid+sp, paid+mp could have interesting results, perhaps even with microtransaction data
- Could be worthwhile to rework recommendation system for multiplayer games to certain types of players (combine this data with player activity data)



Thank You!
